Business Statistics and Insights

Group Project – Data-Driven Business Decision Making

Superstore Sales

The objective of this project is not to produce statistics. The objective is to transform data into evidence, evidence into insights, and insights into decisions. Successful projects will demonstrate not only technical competence but also the ability to think critically, communicate effectively, and support business decisions through data. Use and apply the concepts that you have learned during the course.

Superstore Sales Dataset

available on Kaggle:

https://www.kaggle.com/datasets/vivek468/superstore-dataset-final

The dataset contains information on sales transactions, including:

- Sales
- Profit
- Quantity sold
- Discounts
- Product categories and sub-categories
- Customer segments
- Regions
- Shipping modes
- Order dates and shipping dates

The dataset combines numerical, categorical and time-related variables, making it suitable for exploratory analysis, statistical inference, and business decision-making.

Note: You are not expected to simply calculate statistics. Instead, you should use data to understand a business problem, support decisions, and communicate insights effectively.

The project is organized around four stages:

1. **Data Preparation**
2. **Descriptive Statistics**
3. **Statistical Inference**
4. **Recommendations and Business Insights**

Task 1 – Data Preparation

Before any analysis is performed, you must assess the quality of the dataset and prepare it for analysis.

Your report should address the following questions:

Dataset Understanding

- How many observations and variables are included?
- What types of variables exist?
    - Numerical
    - Categorical
    - Date variables

Data Quality Assessment

Investigate:

- Missing values
- Duplicate records
- Inconsistent categories
- Extreme observations (outliers)
- Data type issues

Variable Transformation

Consider whether it is useful to create new variables such as:

- Profit Margin
- Delivery Time
- Revenue per Order
- Discount Categories

- Year, Quarter and Month

You should justify all cleaning and transformation decisions.

Task 2 – Descriptive Statistics

The goal of this stage is to understand the characteristics of the business.

Central Tendency

Compute and interpret:

- Mean
- Median
- Mode

for relevant variables such as:

- Sales
- Profit
- Quantity
- Discount

Dispersion

Compute and interpret:

- Range
- Variance
- Standard Deviation
- Interquartile Range

Shape of Distributions

Discuss:

- Skewness
- Concentration
- Extreme values

Comparative Analysis

Compare relevant groups, such as:

- Regions
- Customer Segments
- Product Categories
- Shipping Modes

The emphasis should be on interpretation rather than computation.

Task 3 – Statistical Inference

Use inferential methods to support or challenge business assumptions.

Possible techniques include:

Confidence Intervals

Estimate intervals for:

- Average sales
- Average profit
- Average delivery time

Comparison of Means

Examples:

- Profit by region
- Sales by customer segment
- Delivery time by shipping mode

ANOVA

Compare multiple groups simultaneously.

Chi-Square Tests

Study relationships between categorical variables.

The objective is not only to perform the tests but also to explain their business implications.

Task 4 – Recommendations and Business Insights

This is the most important component of the project.

You must transform statistical findings into actionable recommendations.

Your report should answer:

What does the data suggest about business performance, customer behaviour and managerial decision-making?

Your discussion should:

- Connect findings logically;
- Prioritize business relevance;
- Avoid excessive technical language;
- Support managerial decisions with evidence.

Team Topics

Each team will investigate a different business problem using the same dataset.

Team 1 – Product Profitability

Business Question

Why do some products generate losses despite producing sales?

Possible Topics

- Sales versus profit
- Discount strategies
- Loss-making products
- Product portfolio management

Final Recommendation

How should management optimize the product mix?

Team 2 – Regional Performance

Business Question

Which regions perform best and why?

Possible Topics

- Sales by region
- Profit by region
- Regional differences
- Market opportunities

Final Recommendation

How should resources be allocated across regions?

Team 3 – Discounts and Profitability

Business Question

Do discounts increase business performance or reduce profitability?

Possible Topics

- Discount distribution
- Profit margins
- Sales growth versus profitability

Final Recommendation

How should discount policies be redesigned?

Team 4 – Customer Segments

Business Question

Which customer segments create the most value?

Possible Topics

- Sales by segment
- Profit by segment
- Purchasing behaviour

Final Recommendation

Which segments should receive greater marketing attention?

Team 5 – Operations and Shipping

Business Question

How efficient is the delivery process?

Possible Topics

- Delivery times
- Shipping modes
- Operational performance

Final Recommendation

How can logistics be improved?

Team 6 – Sales Trends and Business Growth

Business Question

What patterns emerge over time?

Possible Topics

- Monthly sales
- Seasonal effects
- Growth patterns

Final Recommendation

How should planning and forecasting be improved?

Deliverables

Each team must submit:

Written Report

Maximum: 15 pages

Suggested structure:

1. Introduction
2. Business Problem
3. Data Preparation
4. Descriptive Statistics
5. Statistical Inference
6. Visualizations
7. Recommendations
8. Conclusions

Presentation

Maximum: 10 minutes

The final slide must answer:

If you were the CEO, what three actions would you implement tomorrow based on the data?

Assessment Criteria

Component	**Weight**
Data Preparation	20%
Descriptive Statistics	20%
Statistical Inference	20%
Visualizations and Communication	15%
Business Recommendations	20%
Overall Quality and Professionalism	5%